

## **ENEL 300 Project: Radio-Controlled Car with Metal Detection and Distance Sensing**

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### Revision Log

| # | Date                          | Description                                     |
|---|-------------------------------|-------------------------------------------------|
| 0 | Jan. 13, 2026                 | Original                                        |
| 1 | <a href="#">Mar. 11, 2026</a> | <a href="#">Fixed Fig. 1a and Section A.1.2</a> |

# 1 Project Overview

In this course, you are tasked with designing a radio-controlled car to compete in a final competition at the end of the semester. Additional details for the competition can be found in Appendix A of this document.

## 1.1 Motivation for project

This project is specifically designed to combine concepts that you will be learning in ENEL 343, ENEL 327, ENEL 361 and ENEL 369. The aim is to strengthen your understanding of the material in these courses. Additionally, you will gain practical hands-on experience as well as experience with popular software tools such as KiCad for designing PCBs, Fusion360, and either LTspice or QSPICE for simulating circuits. By working on this project, you will be exposed to many interesting topics, including machines, motor driver circuits, RF communication, circuit design, batteries, power budget planning, project management, printed circuit board design, power converter circuits, circuit protection, signal conditioning, embedded systems, 3D printing, and mechanical design, etc.

## 1.2 Technical Requirements

A summary of the technical requirements for the car are as follows:

| Requirement      | Description                                                                                                                                                                                                                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Movement         | <ul style="list-style-type: none"><li>The car must be able to turn and move forward.</li></ul>                                                                                                                                                                                                                                                                          |
| Power            | <ul style="list-style-type: none"><li>The car must be self-powered, i.e. via a battery.</li></ul>                                                                                                                                                                                                                                                                       |
| Metal Detection  | <ul style="list-style-type: none"><li>The car must be designed to detect copper, in the form of round copper plates that will be attached to the ground and within the boundaries of the circuit. The copper plates are 1 mm thick and have a diameter of 10 cm.</li></ul>                                                                                              |
| Object detection | <ul style="list-style-type: none"><li>The car must be able to determine the distance between it and objects that may be up to 80 cm away from the center of the circuit and at a height of 5cm or greater. The distance measurements will be from designated spots marked with a “X” within the boundaries of the circuit. See the Appendix for more details.</li></ul> |
| Noise Generation | <ul style="list-style-type: none"><li>The car must have the ability to emit a noise in response to detecting the metal.</li></ul>                                                                                                                                                                                                                                       |
| Light Generation | <ul style="list-style-type: none"><li>The car must have a headlight that can be turned on/off.</li></ul>                                                                                                                                                                                                                                                                |
| Wireless Control | <ul style="list-style-type: none"><li>The car must be able to be controlled wirelessly via a remote controller.</li></ul>                                                                                                                                                                                                                                               |
| Controller       | <ul style="list-style-type: none"><li>The remote control must be designed by the team, i.e. cannot be purchased off-the-shelf.</li></ul>                                                                                                                                                                                                                                |
| Display          | <ul style="list-style-type: none"><li>The distance of the object reading needs to be visible to the user.</li></ul>                                                                                                                                                                                                                                                     |

## 1.3 Design Constraints

- Prebuilt modules (i.e. products that contain multiple parts to realize a function) are only allowed for the object/distance detection (e.g. HC-SR04), wireless communication (e.g. HC-05) and for the MCU (e.g. NUCLEO-F446RE). An IC is not considered to be a prebuilt module (i.e. DRV8251ADDAR) as it is a singular part nor is a wheel for your car considered to be a prebuilt module. An example of a prebuilt module that cannot be used include something like this <https://www.amazon.ca/Lithium-Battery-Circuit-Regulator-Protection/dp/B0982DGD45> or this <https://www.amazon.ca/Detector-Module-Electronic-Entertainment-Detection/dp/B0CFX1PW5L>.
- Arduino is not allowed for this project nor is Arduino IDE or Arduino libraries.

3. While we provide resources for STM32 (e.g. NUCLEO-F446RE and the STM32 workshop), other MCUs can be used in your RC car. For instance, you are free to use the AVR board from your ENEL 369 class (AVR128DB28-I/SP) or ESP32, if you prefer.
4. Throughout the semester, it is expected that you will also be using breadboards as you work through your designs. However, the car on the demo day must include at least one custom PCB. Note, a PCB has more component flexibility and improved performance than either a perfboard or breadboard. For instance, regarding component flexibility, in addition to hole components, surface mount components can also be used in PCB. Therefore, it is recommended to embed as much of the desired functionality on your PCB.
5. The maximum allowable cost of the components used in your final car design is \$120 CAD. The spending limit agreed upon by your group must be established in advance and documented in your Team Contract. This \$120 CAD limit applies only to the final design; expenses incurred during prototyping, testing, or debugging throughout the term are not included. You are not required to spend the full \$120 CAD. The amount of money spent on the project will not affect your grade on this course.
6. At least one critical component (i.e. not decorative) for the car must be 3D printed, i.e. chassis, gears, etc.
7. You cannot use an integrated wheel, motor and gearbox combo such as the “TT” DC Gearbox Motor”+ “TT” compatible wheel.

## 1.4 Parts Provided

We will provide each group with **2x NUCLEO-F446RE** boards, **1x HC-05**, and **1x HC-SR04**. These units are on loan and must be returned at the end of the course, even if they are damaged.

If a component becomes non-functional during the term, the course will not provide a replacement. In this event, your group can choose to purchase a replacement yourself (these are standard, widely available, low-cost parts) or adjust your project design accordingly. Any additional materials required for your car should be sourced by your team or through the Electrical Makerspace.

## 2 Team Composition

Each group should have **five members**. You are free to form your own groups. If you would prefer to be assigned a group, or if you are having difficulty forming a group, please notify the appropriate TA.

## 3 Deliverables

As per the course outline, this project will feature seven sprints. There will also be a final technical report due at the end of the term. Here is a brief overview of the individual sprints.

| Sprint | Due Date | Description                                                                                                                                                                                            |
|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Feb. 11  | <b>Initial Design &amp; Planning:</b> Define groups, team responsibilities, and present the initial design and project plan.                                                                           |
| 2      | Feb. 25  | <b>Subsystem 1 - Mechanical:</b> Focus on the structural assembly of the car, including chassis integration and motor mounting. <b>Deliverable:</b> Demonstrate turning and forward/backward movement. |
| 3      | Mar. 4   | <b>Subsystem 2 - Metal Detection:</b> Verification and testing of the metal detector sensing circuits and integration.                                                                                 |

|   |         |                                                                                                                                       |
|---|---------|---------------------------------------------------------------------------------------------------------------------------------------|
| 4 | Mar. 11 | <b>Subsystem 3 - Object &amp; Distance Sensing:</b> Demonstration of distance sensing capabilities.                                   |
| 5 | Mar. 18 | <b>Subsystem 4 - Remote Control:</b> Demonstrate the ability to control the car wirelessly via the remote-control interface.          |
| 6 | Apr. 1  | <b>Minimum Viable Product (MVP):</b> Integration of all subsystems into a functional prototype that meets basic project requirements. |
| 7 | Apr. 15 | <b>Final Demo:</b> Formal presentation and demonstration of the fully optimized car completing the final course challenge.            |

Note, please refer to D2L for the due dates for each of the sprints and for a more detailed explanation of what is expected for each of the sprints.

## 4 Grades

The final grades for the course will be distributed as follows:

| Component                              | Weight |
|----------------------------------------|--------|
| Sprint Deliverables                    | 84%    |
| Technical Workshops                    | 3%     |
| High Performance Teamwork Deliverables | 3%     |
| Final Report                           | 10%    |

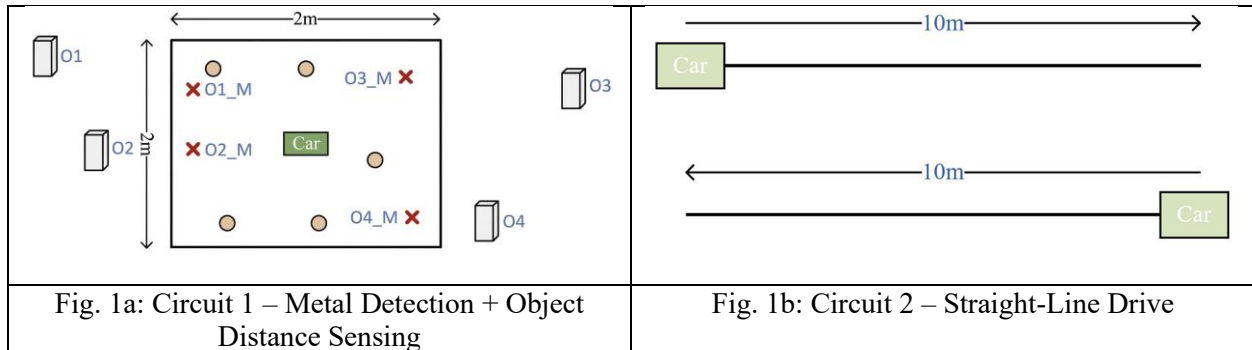
\*Note, the instructor has discretion to award up to a 5% bonus for outstanding work at the end of the term.

The rubrics and the grading scheme for the individual components will be specified later.

Most assessments will be graded on a group basis. Because of the project's complexity, we understand that individual team members may focus on different parts of the work and therefore might not contribute equally to each deliverable. However, by the end of the project, we expect each member's overall contribution to be roughly equal. If you find that a teammate is not contributing adequately at any point, please discuss with an instructor so we can address the situation.

## A Appendix A: Competition Details

In this competition, you will be tasked with completing the two circuits that are graphically illustrated as follows:



### A.1 Overview of Circuit 1

#### A.1.1 Structure

The boundary of the circuit is 2 m x 2 m. Inside the boundary of the circuit, there will be copper plates on the ground. Outside the boundary of the circuit, there will be physical objects, with heights of at least 5 cm. The physical objects could be as far as 0.8m away from the boundary of the circuit.

#### A.1.2 Task #1 – Determine Distance to Object

The car will be tasked with determining the distance of each physical object from specific locations. These locations will be marked with a “X” in the circuit, as shown in Fig. 1a. **The total distance from the designated spot and the object will be less than 80cm. The measurements should be accurate to within  $\pm 2.5$  cm for distances less than 60 cm and  $\pm 3.5$  cm for distances less than 80 cm.** The results must be displayed to the examiner during the task. The display could be located on the car itself, the remote controller, a computer or phone.

#### A.1.3 Task #2– Metal Detection

The car will be tasked with detecting each copper plate within the boundary of the circuit. When copper has been detected, an audible sound must be emitted. Otherwise, no audible noise should be emitted.

#### A.1.4 Procedure

The car will start from the edge of the circuit. The timer will then start. The timer will stop after the team says “stop”.

## A.2 Overview of Circuit 2:

### A.2.1 Structure

This is an open circuit with a defined start and end point that are 10 m apart. This is shown in Fig. 1b. The car is freely able to move between these start and end points. There will also be a ramp between the start and end points. A sketch of the ramp is shown in the following Figure:

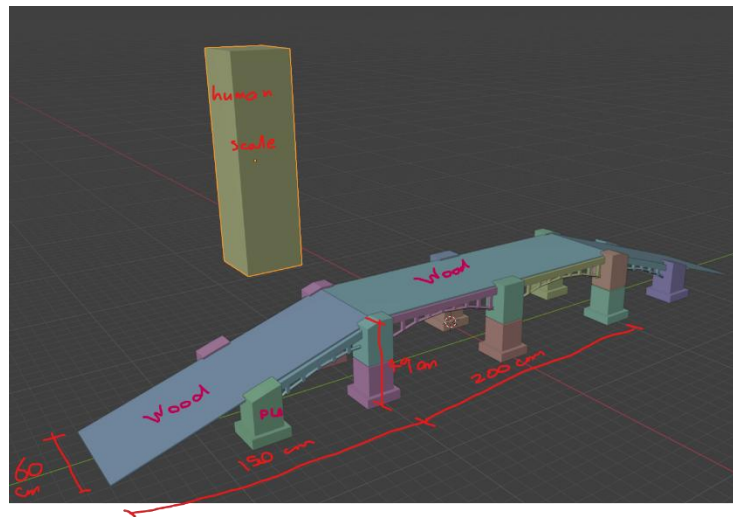


Fig. 2 – Sketch of the Competition Ramp

The ramp is made from wood and is approximately 6 m long and is approximately 60 cm in width. The incline is approximately 20 degrees with a height of approximately 50 cm.

### A.2.2 Task

The car will be tasked with first driving from the start to the end point and then back to the start point.

### A.2.3 Procedure

The timer will stop after the car returns to the starting point.